

Encoder-Decoder LSTM for Sequence-to-Sequence Prediction

Learn to reverse the first three elements of a six-element integer sequence

1. DATA GENERATION & PREPROCESSING

1.1 Generate Source Sequence (Length = 6)

Random integers from 1 to 50
(0 is reserved as start token)

Example:
[13, 28, 18, 7, 9, 5]

1.2 Create Target Sequence (Length = 3)

Take first three elements
and reverse the order

Example:
[18, 28, 13]

1.3 Create Shifted Decoder Input

Add start token (0) and
remove last element

Example:
[0, 18, 28]

1.4 One-Hot Encode Sequences

Convert each integer to a
51-dimensional one-hot vector
(50 values + 1 start token)

Shapes:

X1 (source): (samples, 6, 51)
X2 (decoder in): (samples, 3, 51)
y (target): (samples, 3, 51)

Example After Encoding

X1 (source) -> (samples, 6, 51)
X2 (decoder) -> (samples, 3, 51)
y (target) -> (samples, 3, 51)

2. MODEL ARCHITECTURE

2.1 Training Model (Teacher Forcing)

Encoder LSTM
128 units

Input:
X1
(None, 6, 51)

LSTM
128 units
return_state=True

Outputs:
Hidden State (h)
Cell State (c)
Shapes: (None, 128)

Decoder LSTM
128 units

Input:
X2
(None, 3, 51)

LSTM
128 units
return_sequences=True
return_state=True

Dense
51 units
activation=softmax

Output:
Predicted y
(None, 3, 51)

Initial
State
(h, c)

2.2 Compile & Train

Optimizer: Adam
Loss: Categorical Crossentropy
Metric: Accuracy
Training Data: 10,000 sequences
Batch Size: 64
Epochs: 20
Validation Split: 20%

2.3 Trained Weights

The model learns to map:
Six integers
First three in -> reverse order

3. INFERENCE (PREDICTION)

3.1 Inference Encoder

Input: New source sequence X1
(1, 6, 51)

Encoder LSTM
(128 units)

Outputs: Final states
Hidden State (h)
Cell State (c)

3.2 Inference Decoder (Step-by-Step)

Generate output one element at a time

Start with
<START> token
(0)

1 0 0 0 ... 0

(1, 1, 51)

Decoder LSTM
(128 units)

Dense + Softmax
(51 units)

Output: Next element
(1, 1, 51)

Update states (h, c)
and use prediction as
next input

Generated
3 elements?

Yes

No

Output: Predicted sequence
(3 integers)

4. EVALUATION & OUTPUTS

4.1 Evaluate on Unseen Data

Generate 100 new sequences
Compare predicted sequence (yhat)
to expected sequence (y)
Require ALL 3 elements to match
for a correct prediction

Exact Sequence Accuracy

Correct / 100 x 100%
Example Result:
96/100 (96.00%)

4.2 Display & Save Predictions

Show 10 example predictions
in human-readable form
Save all results to:
runtime_predictions.txt

Example Output Format

X= [22, 17, 23, 5, 29, 11]
y= [23, 17, 22]
yhat=[23, 17, 22]
...

LEGEND

Data Preparation

Model Components

Dense/Output Layer

States / Information

Inference & Evaluation

Data Flow

Loop / Repeat

What the Model Learns

Given a sequence of six integers [a1, a2, a3, a4, a5, a6],
the model learns to output [a3, a2, a1]

Key Takeaways

- Encoder summarizes the entire input sequence into states (h, c).
- Decoder generates the output sequence one element at a time.
- The model is evaluated using exact sequence match accuracy.